FINITENESS OF COMPACTLY SUPPORTED COHOMOLOGY FOR CURVES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Vanishing and finiteness for curves with coefficients (Etale Cohomology, Theorem etale-cohomology-theorem-vanishing-affine-curves-coefficients, canonical tag 0GJI). cited at proof lines 4356-4357 and 4362. The theorem whose application yields the finiteness of H^q_c.
- Compactification description of compact support (Lemma lemma-compactify-compact-support, canonical tag 0F75). cited at proof lines 4361. Gives H^q_c(X, F) = H^q(\overline{X}, j_!F) for the compactification.

Source reference. Original theorem 0GJI: sources/stacks-chow/tags.txt, source lines 20653–20653. Cited at proof lines 4356-4357 and 4362: the theorem whose application yields the finiteness of H^q_c.

Lemma 0.2. *Let k be an algebraically closed field. Let X be a separated scheme of finite type type over k of dimension ≤ 1. Let Λ be a Noetherian ring. Let $\mathcal{F}$ be a constructible sheaf of Λ-modules on X which is torsion. Then $H^q_c(X, \mathcal{F})$ is a finite Λ-module.*

Proof. This is a consequence of Étale Cohomology, Theorem 0GJI. Namely, choose a compactification $j : X \to \overline{X}$. After replacing $\overline{X}$ by the scheme theoretic closure of X, we see that we may assume $\dim(\overline{X}) \leq 1$. Then $H^q_c(X, \mathcal{F}) = H^q(\overline{X}, j_!\mathcal{F})$ and the theorem applies. □

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